

UNIT 02

ATOMIC STRUCTURE



The atom consists of three subatomic particles: electrons, protons, and neutrons.

Electrons are negatively charged particles, protons are positively charged particles, and neutrons have no electrical charge. Most of the mass of an atom is located in the nucleus, which contains the protons and neutrons. Electrons orbit the nucleus in energy levels.

BASIC DISCOVERIES

- ▶ **440 BC:** Democritus proposed the idea of atoms as the indivisible units of matter.
- ▶ **1803:** John Dalton proposed his atomic theory, which stated that all matter is made up of atoms, which are indivisible and indestructible.
- ▶ **1897:** J.J. Thomson discovered the electron and proposed the plum pudding model of the atom.
- ▶ **1909:** The charge of an electron, approximately -1.602×10^{-19} coulombs, was discovered by Robert A. Millikan through his famous oil-drop experiment.
- ▶ **1911:** Ernest Rutherford conducted the gold foil experiment and proposed the nuclear model of the atom.
- ▶ **1913:** Niels Bohr proposed the Bohr's model of the atom, which explained the energy levels of electrons.
- ▶ **1932:** James Chadwick discovered the neutron.

SUB-ATOMIC PARTICLES AND THEIR CHARACTERISTICS

The atom consists of three subatomic particles: electrons, protons, and neutrons.

- ▶ Electrons are negatively charged particles, protons are positively charged particles, and neutrons have no electrical charge.
- ▶ Most of the mass of an atom is located in the nucleus, which contains the protons and neutrons. Except protium nuclei of all other atoms contain neutrons.
- ▶ Electrons orbit the nucleus in energy levels.

BASIC DISCOVERIES OF ATOMIC STRUCTURE

ELECTRONS

- ▶ Discovered by J.J. Thomson in 1897.
- ▶ Revolve in orbits around the nucleus.
- ▶ Carry a negative charge equal to the positive charge of a proton.
- ▶ Form less than one percent of the total mass of an atom.
- ▶ The mass of electron is nearly $1/1836$ times less than proton and $1/1839$ times less than neutron.

PROTONS

- ▶ Discovered by Eugen Goldstein in 1886.
- ▶ Located in the nucleus of all atoms.
- ▶ Carry a positive charge equal to the negative charge of an electron.
- ▶ Determine the atomic number of an element.
- ▶ Almost 99.94% mass of an atom exists in the nucleus.

NEUTRONS

Discovered by James Chadwick in 1932.

- ▶ Located in the nucleus of all atoms except hydrogen.
- ▶ Have no charge.
- ▶ Determine the mass number of an element.
- ▶ (Mass number = Number of Protons – Number of neutrons)



BOHR'S ATOMIC MODEL

INTRODUCTION

1913, Bohr gave his Atomic Model with some modifications in Rutherford's Model.

1. Electron can revolve only in certain stable Orbits.
2. These orbits have fixed energy & these are Called Energy Levels. These were named K, L, M, N Corresponding $n = 1, 2, 3, 4 \dots$ and so on. Where 'n' is the principal quantum number.
3. Electron have same energy as that orbit in which it is revolving the or Stationary State.
4. While revolving in a particular Energy Level (orbit) electron do not emit any radiation.
5. The energy is emitted or absorbed only when an electron jumps from one orbit to another.
6. When an electron jumps from lower orbit to higher orbit, it absorbs energy and when it jumps from higher orbit to lower orbit it radiates energy. This change in energy, ΔE is given by following Planck's equation:

$$\Delta E = E_2 - E_1 = h\nu$$

7. Where, 'h' is Planck's constant equal to 6.63×10^{-34} J.s, and 'v' is frequency of light. Electron can revolve only in orbits of a fixed angular momentum (mvr), given as: Angular momentum = $mvr = nh/2\pi$ Where 'n' is the quantum number or orbit number having values 1, 2, 3 and so on.

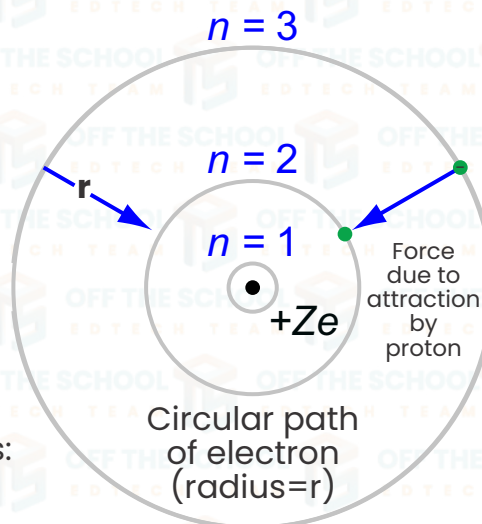
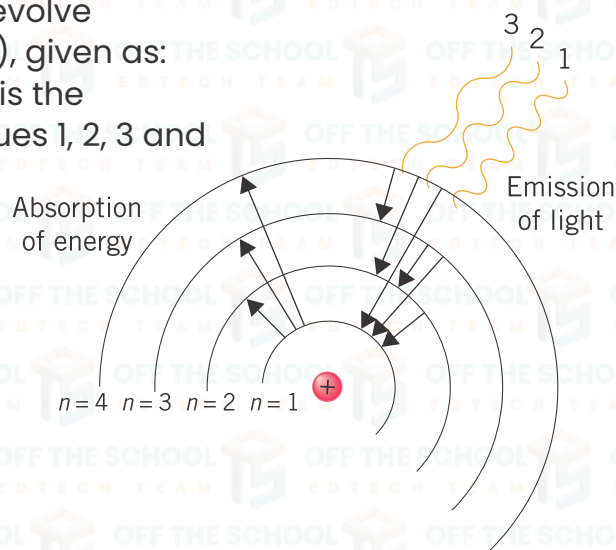
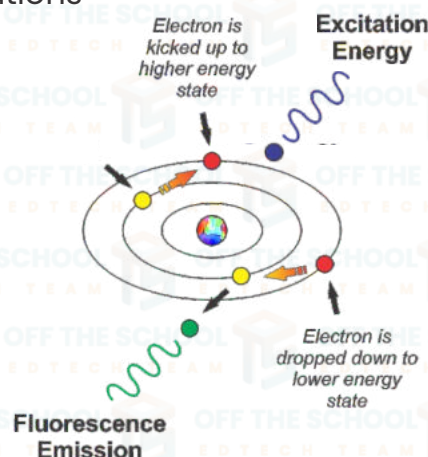
LIMITATIONS OF BOHR'S ATOMIC MODEL

Bohr's model failed to explain:

- ▶ Zeeman Effect.
- ▶ Stark Effect.
- ▶ Heisenberg's uncertainty principle.
- ▶ Spectra of large atoms.
- ▶ Monoelectronic species like H^+ , Li^{+2} etc.

RADIUS OF HYDROGEN ATOM

Let, "m" be the mass of the electron, "e" be the charge on the electron, "v" is the velocity with which the electron is moving in an orbit having radius "r" & "ze" be the positive charge on the nucleus. According to coulomb law, the Electrostatic force of attraction between the nucleus & valance electron can be written as:



$$F_e = \frac{k(q_1 q_2)}{r^2} \longrightarrow \text{eq. (i)}$$

$$F_c = \frac{mv^2}{r} \quad \text{eq. (ii)}$$

$$k = \frac{1}{(4\pi\epsilon_0)}$$

$$q_1 = Ze^+$$

$$q_2 = e^-$$

$$F_e = k \frac{q_1 q_2}{r^2}$$

Putting the value of k, q_1 & q_2 ;

$$F_e = \frac{Ze^2}{4\pi\epsilon_0 r^2} \longrightarrow \text{eq. (iii)}$$

$$F_e = F_c$$

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$$

$$r = \frac{Ze^2}{(4\pi\epsilon_0 mv^2)} \longrightarrow \text{eq. iv}$$

According to Quantized Condition:

$$mvr = \frac{nh}{(2\pi)}$$

$$v = \frac{nh}{2\pi mr} \longrightarrow \text{eq. v}$$

Put from eq. v to eq. iv

$$r = \frac{Ze^2}{[4\pi\epsilon_0 m \left(\frac{nh}{2\pi mr}\right)^2]}$$

$$r = \frac{Ze^2 \times 4\pi^2 m^2 r^2}{4\pi\epsilon_0 m \times n^2 h^2}$$

$$r = \frac{Ze^2 \pi m r^2}{\epsilon_0 (nh)^2}$$

Rearrange this equation for

$$r = \frac{\epsilon_0 (nh)^2}{(Ze^2 \pi m)}$$

$$1 \times 10^{-10} = 1 \text{ \AA (Angstrom).}$$

DERIVATION FOR ENERGY

In an orbit of an atom total energy is the sum of kinetic and potential energy.

$$E = K.E + P.E$$

FOR KINETIC ENERGY

Kinetic energy is due to the motions of electrons in an orbit for which centrifugal force and electrostatic force may be equal.

$$\begin{aligned} F_c &= F_e \\ -\frac{mv^2}{r} &= -\frac{Ze^2}{4\pi\epsilon_0 r^2} \\ mv^2 &= \frac{Ze^2}{4\pi\epsilon_0 r} \end{aligned}$$

Multiply $\frac{1}{2}$ on both sides;

$$\begin{aligned} \frac{1}{2} mv^2 &= \frac{\frac{1}{2} Ze^2}{4\pi\epsilon_0 r} \\ \therefore K.E &= \frac{mv^2}{2} \\ K.E &= \frac{Ze^2}{8\pi\epsilon_0 r} \end{aligned}$$



FOR POTENTIAL ENERGY

Potential energy is the work done in bringing an electron from infinity to distance "r" from the nucleus.

$$P.E = -\text{workdone} = F.e.S \cos\theta$$

$$P.E = -\frac{Ze^2}{4\pi\epsilon_0 r} \times r \cos(0^\circ)$$

$$P.E = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

Here, the negative signs show that P.E decreases (due to attraction b/w nucleus & electron).

$$E = K.E + P.E$$

$$E = \frac{Ze^2}{8\pi\epsilon_0 r} + \left(-\frac{Ze^2}{4\pi\epsilon_0 r}\right)$$

$$E = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$E = \frac{Ze^2}{\pi\epsilon_0 r} \left(\frac{1}{8} - \frac{1}{4}\right)$$

$$E = -\frac{Ze^2}{8\pi\epsilon_0 r}$$

$$E = -\frac{Ze^2}{8\pi\epsilon_0 r} \longrightarrow \text{eq. (iii)}$$

According to Bohr's:

$$r = \frac{\epsilon_0 (nh)^2}{(Ze^2 \pi m)}$$

Put this in eq. (iii)

$$E = -\frac{Ze^2}{8\pi\epsilon_0} \times \frac{\pi m Ze^2}{\epsilon_0 n^2 h^2}$$

$$E = \frac{-Z^2 e^4 m}{8\epsilon_0^2 n^2 h^2} \quad [Z=1]$$

$$E = \frac{-m e^4}{8\epsilon_0^2 n^2 h^2}$$

$$E = \frac{-m e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right)$$

$$\therefore E = -\frac{k}{n^2}$$

EXPRESSION FOR FREQUENCY

When hydrogen atom's electron jumps from a higher level to a lower energy level, a definite amount of energy is emitted.

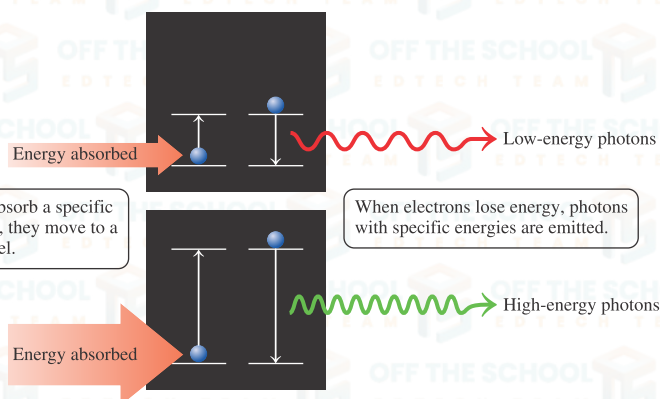
$$\Delta E = E_2 - E_1$$

$$E_1 = \frac{-Z^2 e^4 m}{8\epsilon_0^2 n_1^2 h^2}$$

$$E_2 = \frac{-Z^2 e^4 m}{8\epsilon_0^2 n_2^2 h^2}$$

$$\Delta E = \left(\frac{-Z^2 e^4 m}{8\epsilon_0^2 n_2^2 h^2}\right) - \left(\frac{-Z^2 e^4 m}{8\epsilon_0^2 n_1^2 h^2}\right)$$

$$\Delta E = \left(\frac{-Z^2 e^4 m}{8\epsilon_0^2 n_2^2 h^2}\right) + \left(\frac{Z^2 e^4 m}{8\epsilon_0^2 n_1^2 h^2}\right)$$



$$\Delta E = \frac{Z^2 e^4 m}{8\epsilon_0^2 h^2} \left(\frac{-1}{n_2^2} + \frac{1}{n_1^2}\right)$$

$$\Delta E = \frac{Z^2 e^4 m}{8\epsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

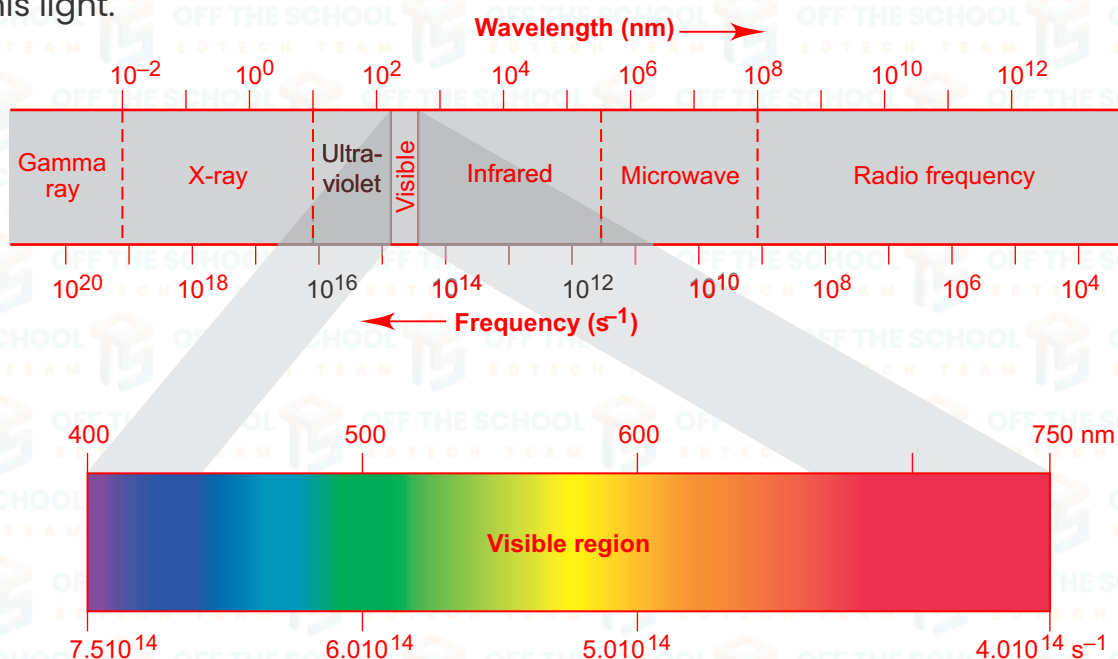
According to Planck's Constant

$$\Delta E = h \nu$$



SPECTRUM

When a polychromatic light is passed through the diffraction grating or prism it split into its components and a band of color is obtained this is the spectrum of this light.

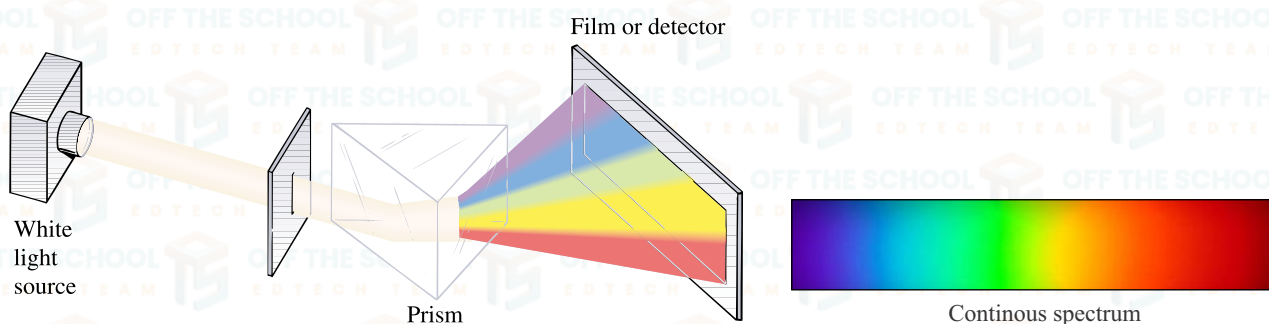


TYPES OF SPECTRUM

1) CONTINUOUS SPECTRUM

The spectrum which contain a continuous band of different color.

EXPLANATION: A continuous spectrum is a spectrum of light that contains an uninterrupted, smooth range of wavelengths, producing a rainbow-like display of colors. It is typically emitted by a hot, dense object or a continuous light source and does not exhibit any distinct lines or gaps.



2) LINE SPECTRUM

A line spectrum is like a special light pattern made up of distinct, separate lines of color. These lines happen because tiny particles in atoms or molecules change their energy levels and give off or take in light in a very organized way.

They are found into two ways:

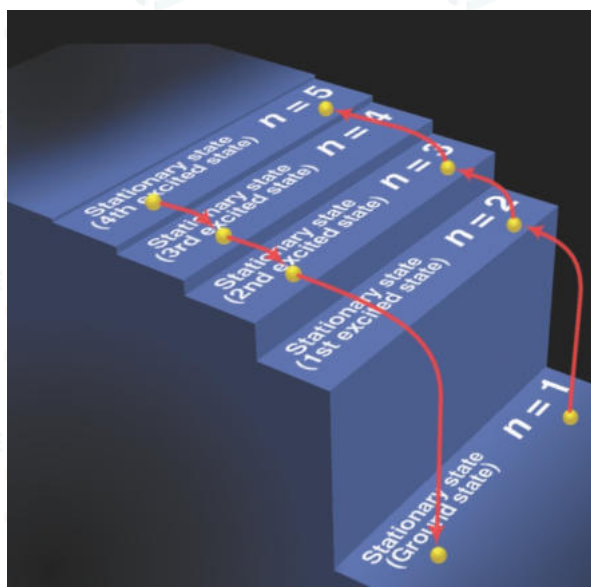
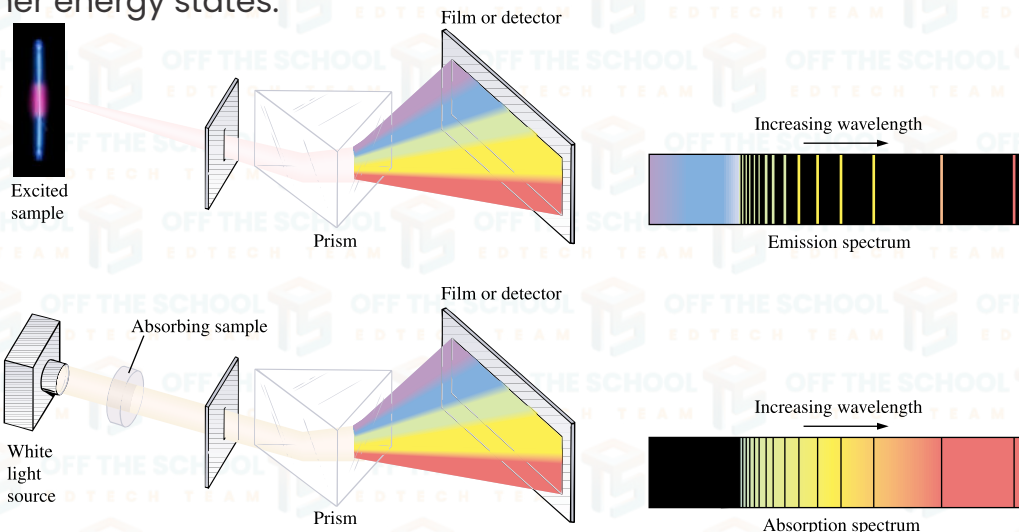
► EMISSION LINE SPECTRUM:

When energy is absorbed by electrons of an atom, electrons move from lower energy levels to higher energy levels. These excited electrons have to radiate energy to return to ground states from the excited state, which is unstable. The emission spectrum is formed by the frequencies of this emitted light.



► ABSORPTION SPECTRUM:

An absorption spectrum is constituted by the frequencies of light transmitted with dark bands when energy is absorbed by the electrons in the ground state to reach higher energy states.



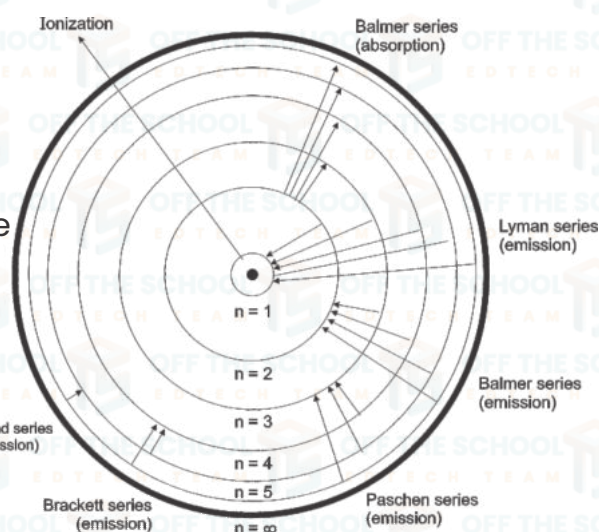
A quantum “staircase” as an analogy for atomic energy levels. Note that the electron can move up or down one or more steps at a time but cannot lie *between* steps.

HYDROGEN SPECTRUM

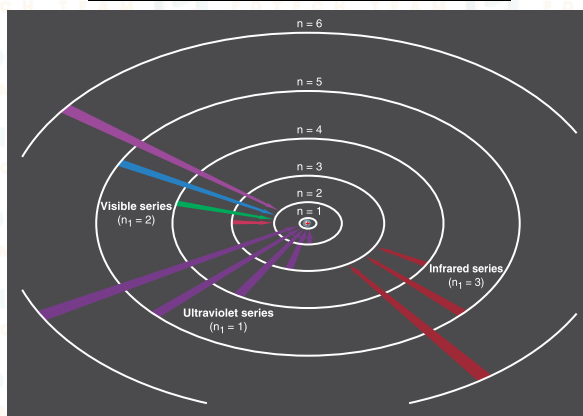
The hydrogen atom produces different patterns of colored lines when it absorbs and releases energy. This happens because electrons in the atom can jump to higher energy levels and then come back down, emitting light of specific colors.

When we pass electricity through hydrogen gas by putting this into discharge tube, it turns into individual hydrogen atoms, and the electrons in these atoms get excited to higher energy levels. However, these excited electrons are unstable and eventually drop back to lower energy levels, releasing energy as light (photons) in the process. This light appears as distinct colored lines.

These colored lines are organized into five series: Lyman, Balmer, Paschen, Brackett, and Pfund. Each series corresponds to electrons returning to specific lower energy levels. The wavelengths of the emitted light can be calculated using a formula called the Balmer equation.

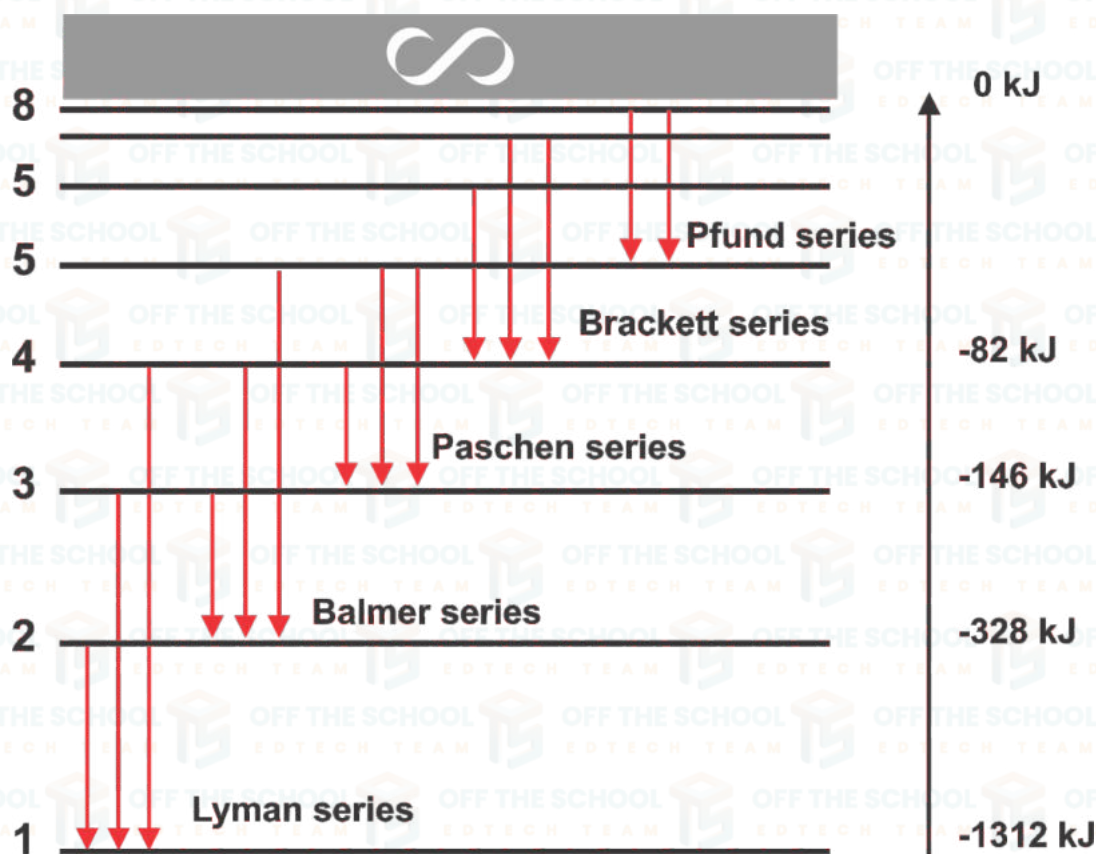


$$\tilde{\nu} = \frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} + \frac{1}{n_2^2} \right]$$



SPECTRAL LINES

Spectral Series	n_1 (Lower energy level)	n_2 (Higher energy level)	Spectral region
1. Lyman	1	2,3,4,... ∞	Ultraviolet
2. Balmer	2	3,4,5,... ∞	Visible
3. Paschen	3	4,5,6,... ∞	Infrared
4. Brackett	4	5,6,7,... ∞	Infrared
5. Pfund	5	6,7,8,... ∞	Far-infrared



PLANCK'S QUANTUM THEORY

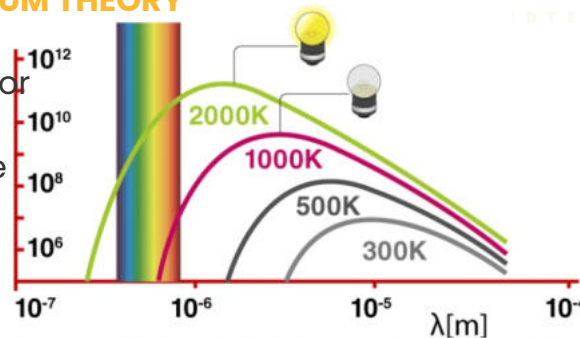
According to Planck's quantum theory,

1. Different atoms and molecules can emit or absorb energy in discrete quantities only. The smallest amount of energy that can be emitted or absorbed in the form of electromagnetic radiation is known as quantum.

2. The energy of the radiation absorbed or emitted is directly proportional to the frequency of the radiation.

$$E \propto \nu$$

$$E = h\nu$$

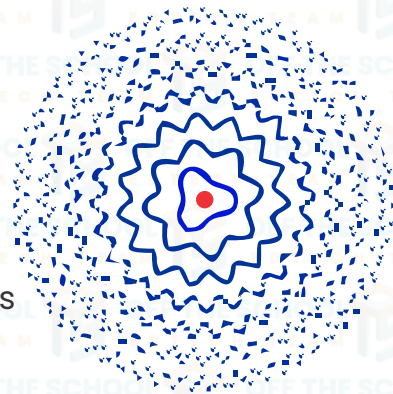


DISCOVERY OF X-RAYS

X-rays were discovered by German physicist Wilhelm Conrad Roentgen on November 8, 1895.

CATEGORIES

The K-series and L-series of X-rays are specific categories of X-ray emissions that result from transitions of electrons within atoms.

**K-SERIES X-RAYS****Schrodinger's atomic model**

- ▶ The K-series X-rays are produced when an electron transitions from a higher energy level (typically $n = 2$) to the innermost energy level ($n = 1$) of an atom.
- ▶ The K-series is further divided into subshells labeled as $K\alpha$ and $K\beta$.
- ▶ $K\alpha$ X-rays result from transitions to the $1s$ subshell ($n=1$) and have slightly higher energy.
- ▶ $K\beta$ X-rays result from transitions to the $2s$ or $2p$ subshells ($n=2$) and have slightly lower energy compared to $K\alpha$ X-rays.
- ▶ These X-rays are characteristic X-rays, meaning they have well-defined and specific energies unique to each element and used in X-ray spectroscopy.

L-SERIES X-RAYS

- ▶ The L-series X-rays are produced when an electron transitions from a higher energy level (typically $n=3$) to the second energy level ($n=2$) of an atom.
- ▶ Like the K-series, the L-series is also divided into subshells, including $L\alpha$, $L\beta$, $L\gamma$, and so on.
- ▶ $L\alpha$ X-rays result from transitions to the $2s$ or $2p$ subshells ($n=2$) and have slightly higher energy.
- ▶ $L\beta$ X-rays result from transitions to the $3s$ or $3p$ subshells ($n=3$) and have slightly lower energy compared to $L\alpha$ X-rays.
- ▶ Similar to the K-series, L-series X-rays are also characteristic X-rays specific to each element, making them valuable for elemental analysis using X-ray spectroscopy.

PRODUCTION OF X-RAYS

X-rays are generated when high-speed electrons, typically produced by an electron gun, are accelerated and focused onto a target material, often made of heavy metal like tungsten. When these high-energy electrons collide with the target material, some of their kinetic energy is converted into X-rays. This process is known as "electron impact."

PROPERTIES OF X-RAYS**1. INVISIBLE:**

X-rays are a form of electromagnetic radiation, but they are invisible to the human eye.

2. PENETRATING:

X-rays can pass through many substances, including soft tissues in the human body, but are blocked by denser materials like bones and metal.

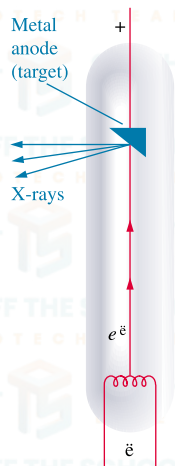
3. HIGH ENERGY:

X-rays have high energy, which allows them to penetrate matter and create detailed images.

4. SPEED OF LIGHT:

X-rays travel at the speed of light and can be focused into a narrow beam.





A simplified representation of the production of X-rays by bombardment of a solid target with a high-energy beam of electrons.

USES OF X-RAYS

1. MEDICAL IMAGING:

X-rays are widely used in medicine for diagnostic imaging, including X-ray radiography and computed tomography (CT) scans to visualize bones, organs, and tissues.

2. DENTISTRY:

Dentists use X-rays to examine teeth and jawbones, helping diagnose dental issues.

3. SECURITY SCREENING:

X-ray machines are used in airports and security checkpoints to scan luggage and detect hidden objects or contraband.

4. SCIENTIFIC RESEARCH:

X-rays are used in scientific research, such as X-ray crystallography to study the atomic and molecular structure of materials.

5. CANCER TREATMENT:

In radiation therapy, X-rays are directed at cancerous tissues to destroy or shrink tumors.

MOSELEY'S LAW

Henry Moseley's most famous statement regarding the relationship between atomic number and X-ray frequency is as follows:

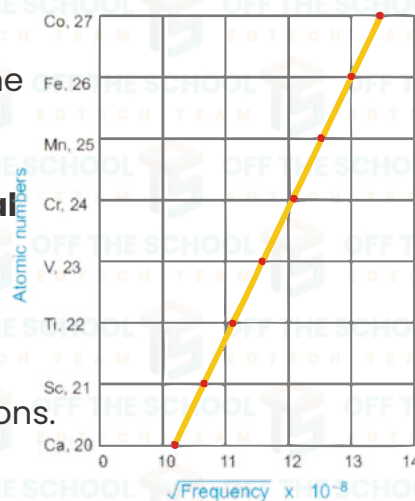
"The square root of frequency is directly proportional to the atomic number (Z) of an element."

Moseley's research showed that there is a direct relationship between the square of an element's atomic number (Z) and the frequency of the X-rays emitted when that element is bombarded with electrons.

$$\sqrt{\nu} \propto (Z-b)$$

$$\sqrt{\nu} = a (Z-b)$$

Where a & b are constants.



RADIOACTIVITY

Discovery of Radioactivity: The discovery of radioactivity is often attributed to Henri Becquerel (1896), a French physicist.

DEFINITION: Continuous emission of invisible radiations from the radioactive substances.

15

DISCOVERY

In 1896, he was studying the phosphorescent properties of uranium salts (Pitchblende). He wrapped a uranium-containing compound in photographic plates and placed them in a drawer with no exposure to light. To his surprise, the photographic plates were still exposed, indicating that the Uranium compound



was emitting radiations even in the absence of light.

Mosley's plot of the square root of X-ray frequencies against atomic number for the elements calcium through cobalt.



NATURAL RADIOACTIVITY

The elements after (Pb-82) that emits radiations spontaneously are known as Natural radioactive elements & phenomena is Natural Radioactivity.

MARIE AND PIERRE CURIE (1898)

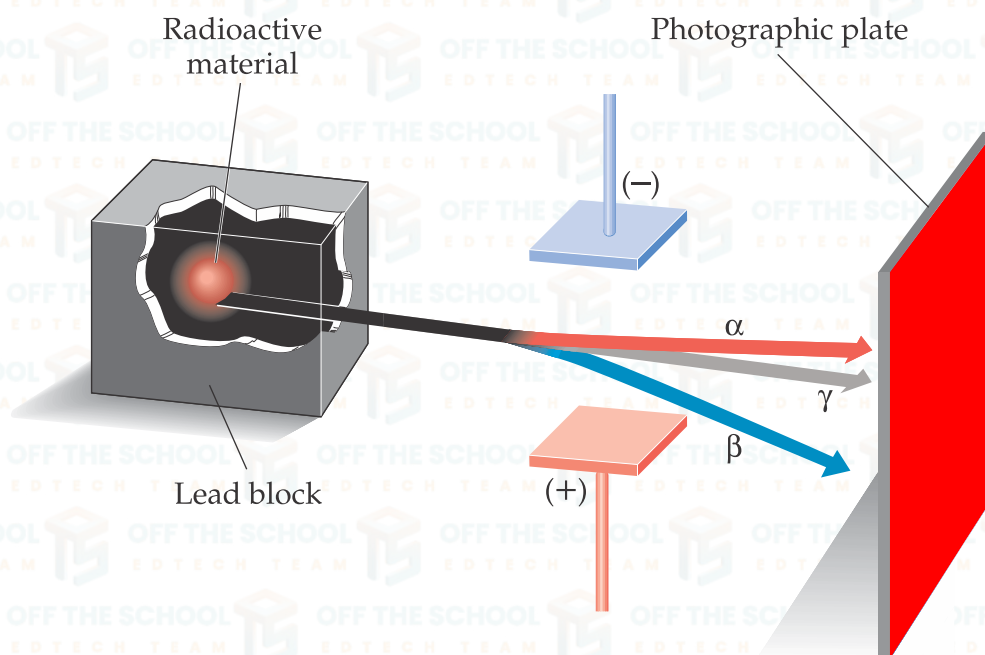
Building on Becquerel's work, Marie Curie and her husband Pierre Curie conducted extensive research on radioactive materials. They discovered the radioactive elements polonium and radium, further solidifying the understanding of radioactivity.

ARTIFICIAL RADIOACTIVITY

The phenomenon in which radiations are produced when the stable element is bombarded with nuclear particles.

TYPES OF RADIOACTIVE RADIATION

Rutherford conducted an experiment and found three radioactive rays:



▲ Behavior of a beam of radioactivity in an electric field.

There are three primary types of radioactive radiation:

ALPHA (α) RADIATION

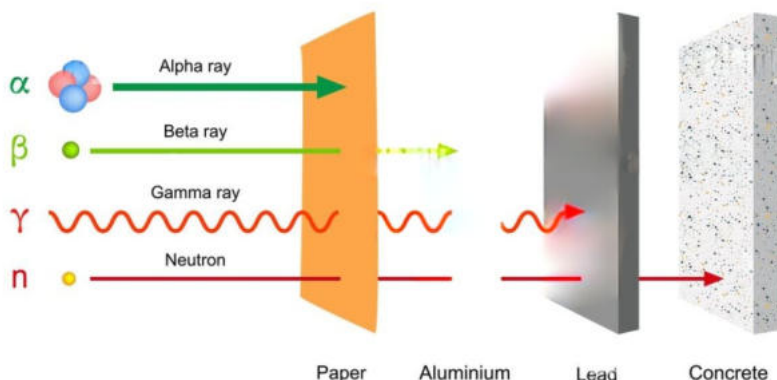
1. Alpha particles consist of two protons and two neutrons, which is essentially a helium nucleus.
2. They have a relatively large mass and a positive charge.
3. Due to their size and charge, alpha particles have low penetration ability and can be stopped by a sheet of paper or human skin.
4. However, if alpha-emitting materials are ingested or inhaled, they can be harmful.

BETA (β) RADIATION

1. Beta particles are high-energy electrons (β^-) or positrons (β^+).
2. Beta- particles have a negative charge, while beta+ particles have a positive charge.
3. They can penetrate materials more deeply than alpha particles but can be stopped by materials like plastic or glass.

GAMMA (γ) RADIATION

1. Gamma rays are high-energy electromagnetic waves.
2. They have no mass and no charge.
3. Gamma radiation is highly penetrating and requires dense materials such as lead or several centimeters of concrete to shield against it.
4. Gamma radiation is often emitted along with alpha or beta radiation.



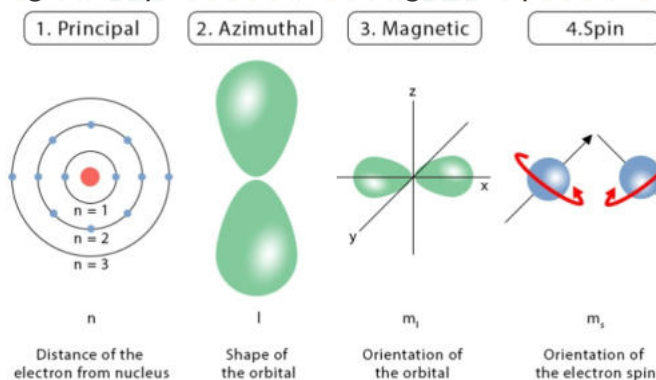
USES OF RADIOACTIVITY

Radioactive materials and radiation have several important applications, including:

1. Medical radiography and radiation therapy for diagnosing and treating diseases.
2. Nuclear power generation.
3. Carbon dating to determine the age of archaeological artifacts.
4. Industrial radiography for testing the integrity of materials.
5. Smoke detectors, which use a small amount of radioactive material to detect smoke.

QUANTUM NUMBERS

This Concept was given by Erwin Schrödinger's equation:



The solution of Schrodinger's wave equation gives certain mathematical integers. These sets of numerical values, which give the acceptable picture of an atom, are called quantum numbers. Consequences of the Quantum Mechanical Treatment of atom are as follows:

- (a) The energy of electrons in atoms is quantized.
- (b) Electrons with different energies are to be found in different regions.
- (e) The position and momentum of an electron cannot be determined at the same time.

DEFINITION

"Mathematical integers derived from the Schrödinger's wave equation is used to describe energy levels, sub-energy levels available for electrons."

OR

"The number or labels which completely describe quantum state of an electron in an atom is called quantum numbers."

TYPES OF QUANTUM NUMBERS

There are four types of quantum numbers which can describes the electron completely.

1) PRINCIPAL QUANTUM NUMBER:

- Denoted by 'n' given by Bohr.
- It describes the orbit or shell or energy levels.
- $n = 1, 2, 3, 4, \dots$ K L M N respectively.
- No. of Subshells in n^{th} orbit = n
- No of orbitals for given value of $n = n^2$.
- Total No. of electrons for a given value of $n = 2n$
- Angular momentum = $mvr = nh/2\pi$

2) AZIMUTHAL QUANTUM NUMBER OR SUBSIDIARY Q.N SECONDARY Q.N OR ORBITAL ANGULAR MOMENTUM Q.N:

- It is denoted by l & given by Sommer field.
- It describes the sub-orbit or Sub-shell or sub-energy level

1	0	1	2	3	4
	s	p	d	f	g

- Azimuthal quantum number ' l ' for any particular orbit is calculated by " $l=0$ to $n-1$ ".

For instance:

Find the value of ' l ' in $n=3$

$$l = 0 \text{ to } n - 1$$

$$l = 0 \text{ to } 3 - 1$$

$$l = 0 \text{ to } 2$$

$$l = 0, 1, 2$$

- It means in 3^{rd} shell there is 3s, 3p, 3d orbitals are exist.
- Maximum number of orbitals for a given value of l is = $(2l+1)$.
- Maximum number of electrons for a given value of l is = $2(2l+1)$.
- **Orbital angular momentum** = $\sqrt{l(l+1)} \times \hbar$, where $\hbar$ is reduced planks constant $h/2\pi$

3) MAGNETIC QUANTUM NUMBER:

- It is denoted by ' m ' or ' m_l ' given by Linde.
- It describes the orientation of orbital.
- The value of ' m '; is $-l$ to $+l$

For instance:

The value of m for:

$$l = 0: -0, +0 = 0$$

$$l = 1: -1, 0, +1$$

$$l = 2: -2, -1, 0, +1, +2$$

now assigning the values:

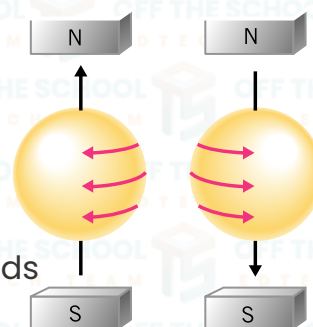
For s orbital ($l = 0, m = 0$):

For P orbital ($l=1, m=1$):

+	0	-1
-1	0	+1
P_x	P_y	P_z
P_y	P_z	P_x

4) SPIN QUANTUM NUMBER:

- It was given in 1925, by Uhlenbeck and Goudsmit.
- It represents the spin of an electron in an orbit.
- It is denoted as "s".
- Its value is $s = \pm \frac{1}{2}$.
- These spins are also designated by arrows pointing upwards and downwards as $\uparrow$ and $\downarrow$.
- Total spin = $\pm n/2$ [where n = no. of unpaired electrons].
- Spin angular momentum = $\sqrt{s(s+1)} (\hbar)$
- $S = n/2$ = no. of unpaired electrons.
- Spin multiplicity = $2s + 1$
- Magnetic moment = $\sqrt{n(n+1)}$ [where n = no. of unpaired electrons]



The electron can spin in either of two directions in the presence of an external magnetic field.

Without a magnetic field the electrons and their spins are usually oriented in random directions.

TABLE 7.1 Quantum Numbers of the Orbitals in the First Four Shells

Value of n	Allowed Value of ℓ	Subshell Label	Allowed Values of m_ℓ	NUMBER OF ORBITALS IN:	
				Subshell	Shell
1	0	s	0	1	1
2	0	s	0	1	4
	1	p	-1, 0, +1	3	
3	0	s	0	1	9
	1	p	-1, 0, +1	3	
	2	d	-2, -1, 0, +1, +2	5	
4	0	s	0	1	16
	1	p	-1, 0, +1	3	
	2	d	-2, -1, 0, +1, +2	5	
	3	f	-3, -2, -1, 0, +1, +2, +3	7	

ORBITALS:

DEFINITION: The region of three-dimensional space around the nucleus where the probability of finding electron is maximum.

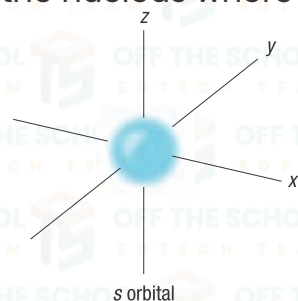
TYPES OF ORBITALS:

s, p, d, and f are the four prominent atomic orbitals.

SHAPES OF ORBITALS:

► S- ORBITAL:

- It has a spherical shape.
- Electrons are uniformly distributed around the nucleus.



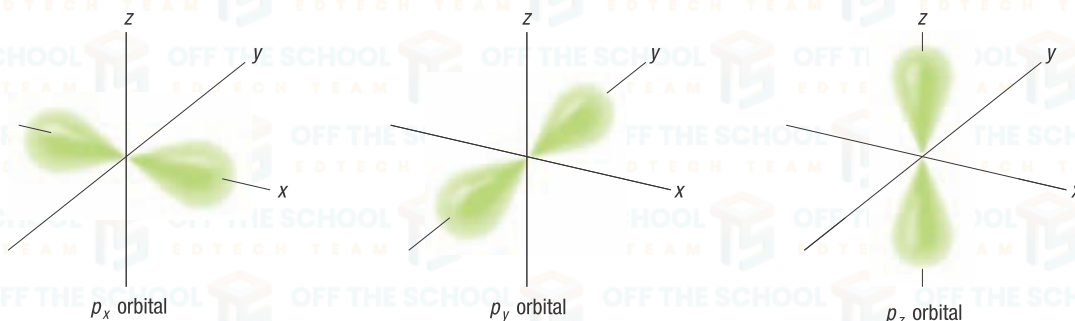
- It has one possible orientation only.
- It has no nodal plane.
- Its size increases with an increase in the value of 'n'.
- The order of size is $1s < 2s < 3s < \dots$

► **P- ORBITALS:**

- It has a dumbbell shape.
- Electrons are distributed around the nucleus in three-dimensional space.
- It has three possible orientations P_x , P_y & P_z These are degenerated orbitals (4) with equal energy.
- It consists of two lobes one above the plane (+) & other below the plane (-) having pear shape.
- It has a nodal plane where two lobes meet each other.
- The size of the p orbitals depends on the principal quantum number n, i.e., $4p > 3p > 2p$.

Three Orientations of p Orbitals

The subscripts x, y, and z indicate the three different orientations of p orbitals. The intersection of the x, y, and z axes indicates the location of the center of the nucleus.

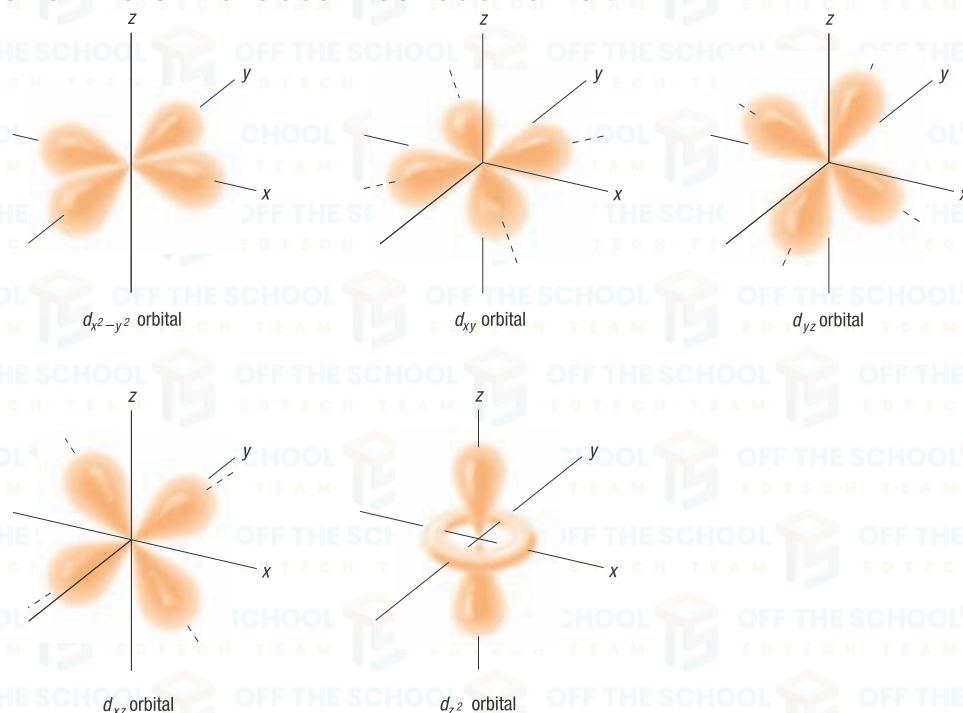


► **D- ORBITAL:**

- It has a double dumbbell shape.
- Electrons are distributed around the nucleus in three-dimensional space. It has five possible orientations d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ & d_{z^2} .
- It has a nodal plane where two lobes meet each other.

five Orientations of d Orbitals

Four of the d orbitals have the same shape but different orientations. The fifth has a different shape and a different orientation. Each orbital occupies a different region of space.



► **F-ORBITALS:**

- It has a saucer shape or complex shape.
- It has seven possible orientations in three-dimensional space.



ELECTRONIC CONFIGURATION:

The representation of filling of electrons in different orbitals of an atom is called electronic configuration.

RULES AND PRINCIPLES OF ELECTRONIC CONFIGURATION:**1. AUFBAU PRINCIPLE:**

GIVEN BY: N. Bohr & Wolfgang Pauli (1925)

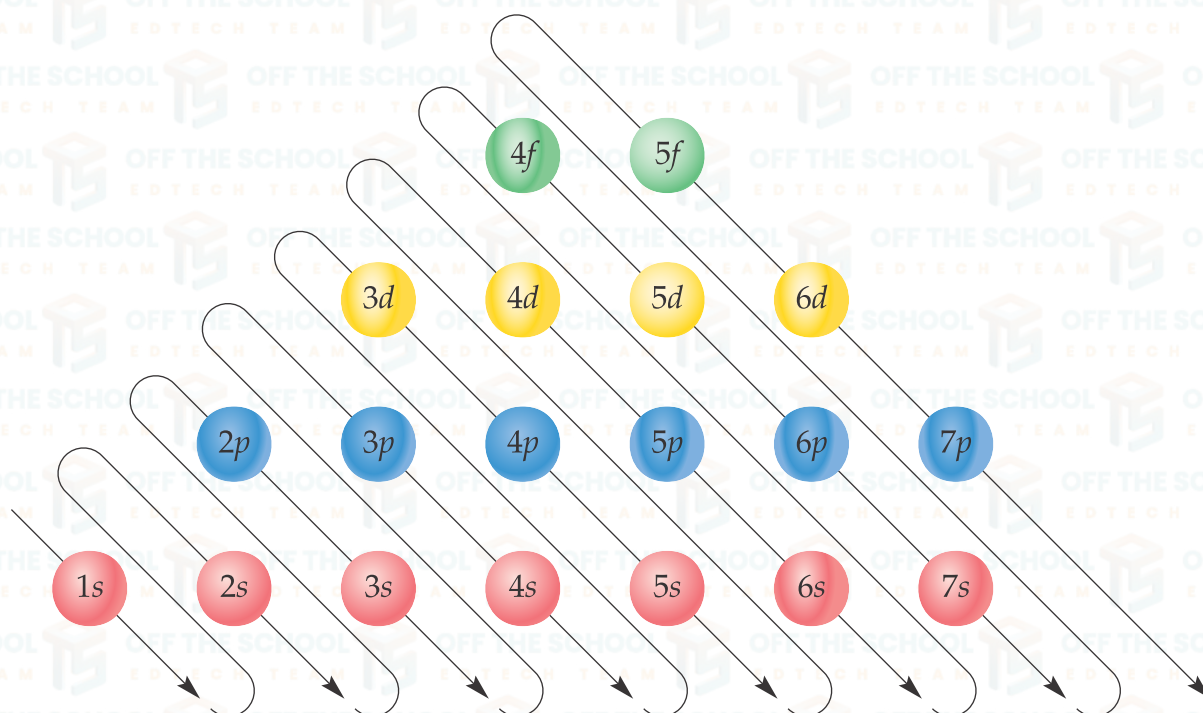
STATEMENT: Electrons are filled progressively to the various subshells in the order of increasing energy, starting with 1s subshell having the lowest energy.

EXPLANATION:

- From this principle, it is clear that electrons are filled in order of increasing energy orbitals.

- The sequence of filling the orbitals is as follows:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p

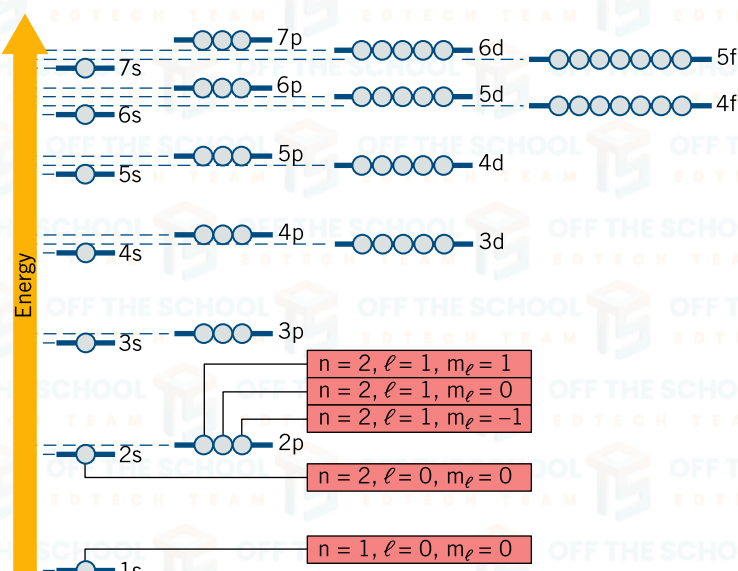


▲ An order-of-filling chart for determining the electronic configuration of atoms.

Approximate energy level diagram for atoms with two or more electrons.

The quantum numbers associated with the orbitals in the first two shells are also shown.

■ In an atom, an electron can take on many different energies and wave shapes, each of which is called an orbital that is identified by a set of values for n , ℓ , and m_ℓ . When the electron wave possesses a given set of n , ℓ , and m_ℓ , we say the electron “occupies the orbital” with that set of quantum numbers.



2. PAULI'S EXCLUSION PRINCIPLE:**GIVEN BY:** Wolfgang Pauli (1925)**STATEMENT:** In an orbital of an atom, no electrons can have the same set of four quantum numbers. At least one quantum number must be different.**EXPLANATION:**

- From this principle, it is found that two electrons have maximum of 3 similar quantum numbers but the fourth quantum number must be different.
- This means that when two electrons are present in the same orbital, they have the same principle, azimuthal & magnetic quantum numbers but the spin quantum number of both electrons must be different.
- That is if one electron has a clockwise spin the other electron would have anti-clockwise spin.

EXAMPLE: Four quantum numbers of electrons present in helium atom, (Z of He = 02) Electronic configuration of He is $1s^2$.

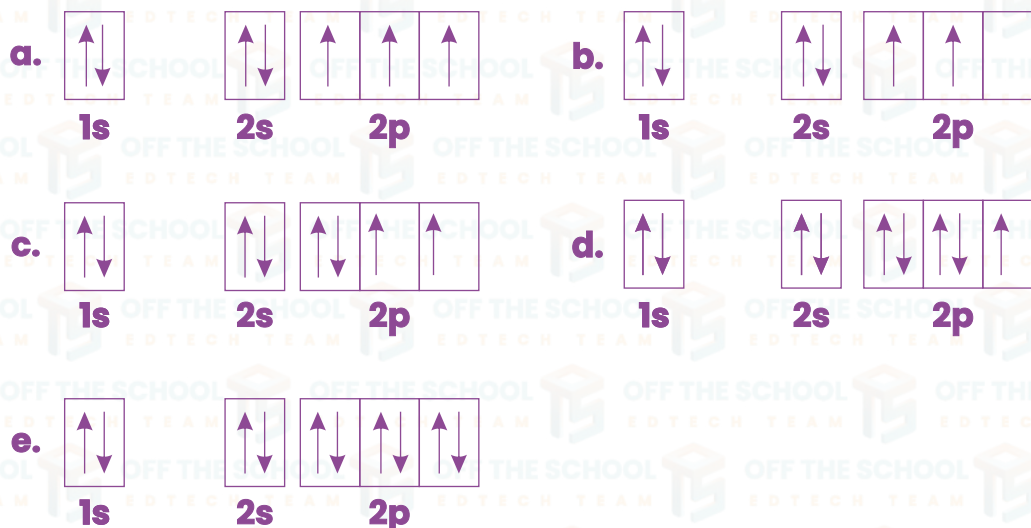
Quantum Numbers	For 1st electron	For 2nd electron
n	1	1
l	0	0
m	0	0
s	$+\frac{1}{2}$	$-\frac{1}{2}$

3. N+L RULE:**GIVEN BY:** Wiswesser**STATEMENT:** Those orbitals which have a lower value of $n + l$ value are filled first, In case two orbitals have the same $n + l$ value, then the orbital having a lower value of n will be filled first.**EXPLANATION:**

Filling of electrons based on $n+l$ rule				
Selection	3d & 4s		5s & 4p	
Orbital	3d	4s	5s	4p
n	3	4	5	4
l	2	0	0	1
$n+l$	5	4	5	5
Preference	The electron goes to 4s		The electron to 4p	

- "n" & "l" are principal & azimuthal quantum numbers respectively.
- This rule tells us that the electrons are filled based on the value of $n+l$, but mainly the filling of electrons is based on the principal quantum number "n". Where there is a tie between the values of " $n+l$ " the orbital with the lower value of "n" gives priority.

4. HUND'S RULE:**GIVEN BY:** Friedrich Hund (1927).**STATEMENT:** In available degenerate orbitals (p, d, f) electrons are distributed in such a way that the maximum number of half-filled orbitals (single electron in an orbit) is obtained.**Correct****Incorrect**



Hund's Rule

A. 1S, 2S, 2P ORBITALS:

- **Configuration:** $1s^2 2s^2 2p^3$
- **Element:** Nitrogen (N)
- **Explanation:** Nitrogen has 7 electrons. The 1s and 2s orbitals are fully occupied, and the three 2p electrons each occupy separate orbitals with parallel spins according to Hund's Rule.

B. 1S, 2S, 2P ORBITALS:

- **Configuration:** $1s^2 2s^2 2p^2$
- **Element:** Carbon (C)
- **Explanation:** Carbon has 6 electrons. The 1s and 2s orbitals are fully occupied, and the two 2p electrons occupy separate orbitals with parallel spins.

C. 1S, 2S, 2P ORBITALS:

- **Configuration:** $1s^2 2s^2 2p^4$
- **Element:** Oxygen (O)
- **Explanation:** Oxygen has 8 electrons. The 1s and 2s orbitals are fully occupied, and the four 2p electrons occupy the orbitals such that one orbital has a pair of electrons and the other two each have one electron.

D. 1S, 2S, 2P ORBITALS:

- **Configuration:** $1s^2 2s^2 2p^5$
- **Element:** Fluorine (F)
- **Explanation:** Fluorine has 9 electrons. The 1s and 2s orbitals are fully occupied, and the five 2p electrons fill the orbitals such that one orbital has a pair of electrons, and the other two each have one electron.

E. 1S, 2S, 2P ORBITALS:

- **Configuration:** $1s^2 2s^2 2p^6$
- **Element:** Neon (Ne)
- **Explanation:** Neon has 10 electrons. The 1s and 2s orbitals are fully occupied, and the six 2p electrons fill all three 2p orbitals, with each orbital having a pair of electrons with opposite spins.

